

Enhancing Dialectal Arabic Sentiment Analysis Using Deep Learning

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This study aims to enhance Arabic Sentiment Analysis (ASA) by developing and evaluating a hybrid Deep Learning (DL) model, AraBERT_CNN_MHA_BiLSTM, which integrates Arabic Bidirectional Encoder Representations from Transformers (AraBERT) with Convolutional Neural Networks (CNN), Multi-Head Attention (MHA), and Bidirectional Long Short-Term Memory (BiLSTM) for Sentiment Analysis (SA) in the Iraqi Arabic dialect. Through an ablation study, we systematically assess the contribution of each component by testing the model on two datasets: "IQAD31K," comprising 31,324 reviews scraped from Google Play Store, and the Iraqi Arabic Dialect (IAD) dataset, comprising 2,000 annotated comments from Iraqi Facebook pages. The complete model achieves an F1-score of 95.63% on "IQAD31K" and 94.50% on the IAD dataset, exceeding traditional Machine Learning (ML) methods such as Support Vector Machine (SVM) and Random Forest (RF), which struggle with Arabic linguistic complications. These findings highlight the importance of integrating AraBERT's contextual embeddings, CNN's local feature extraction, MHA's long-range dependency capture, and BiLSTM's sequential modeling to achieve the best performance in dialectal Arabic sentiment analysis, thereby contributing a robust architectural framework for future research in dialectal Arabic Natural Language Processing (ANLP).

CCS Concepts: • **Information systems** → **Sentiment analysis** • **Computing methodologies** → **Neural networks**; **Machine learning algorithms**;

Additional Key Words and Phrases: Arabic sentiment analysis, iraqi arabic dialect, AraBERT, deep learning, ablation study

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1 Introduction

The Internet has become an essential space for user-generated content, featuring a diverse range of text, images, and emoticons. It covers many areas, such as business, healthcare, politics, education, and food delivery [1]. Every time users log into social networking sites, they create a variety of content. This includes casual text, personal registration details, videos, and other

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forms of digital expression. Content created by users has gained significant attention because it holds great importance for various fields, including businesses, government agencies, and political groups. It is essential to consider the feelings, thoughts, and perspectives expressed in this content to make informed choices. This is especially true when all these insights come together in one place and focus on a specific topic. Understanding public opinion through these gathered comments provides valuable insights, helping stakeholders make informed decisions based on real data in their areas of expertise [2].

The interactions of individuals on Facebook, X (formerly known as Twitter), and Instagram reveal a wide range of experiences and views. For example, users might express anger with a recently acquired product, deliberate on modern societal issues, or state their political viewpoints. The functionality of many applications, including recommendation systems, organizational survey analyses, and strategic planning for political campaigns, depends critically on the data collected on these platforms. For government organizations, public **Sentiment Analysis (SA)** is essential, as it helps clarify human behavior and explains how the interaction of group views shapes such behavior. Moreover, the inference of user sentiment has been quite helpful in the fields of recommender systems and personalization since it provides a means to solve the lack of explicit user feedback about a particular service, thereby improving the accuracy and relevance of customized recommendations [3].

A special subfield of **Natural Language Processing (NLP)**, SA is a branch of computational linguistics that focuses on the computational perspectives of human language. Machine translation, text generation, language comprehension, speech recognition, **named entity recognition (NER)**, **question answering (QA)**, information retrieval, and relationship extraction are just a few of the many uses and research areas NLP spans. Commonly referred to in academic literature as opinion analysis, SA uses NLP, statistical approaches, or **Machine Learning (ML)** to detect the emotional subtleties in text. This procedure involves the automated classification of opinions about specific topics within the text, distinguishing them as either “positive”, “negative”, or “neutral”. SA helps to extract subjective insights by means of methodical textual data analysis, enabling a more complete knowledge of the attitudes and points of view expressed in the material text [4].

Arabic language poses special issues to computational research and is an active field of study. Meanwhile, it is also the fourth most popular language on the Internet [5] and the fifth most spoken language worldwide [6].

Even if SA methods have advanced significantly, the study of Arabic opinions remains limited and is regarded as a challenging task for several reasons. The Arabic language primarily consists of complex syntactic and morphological structures that limit computational capabilities. The limited resources available for **Arabic sentiment analysis (ASA)** limit its application and advancement. Arabic also exhibits notable morphological variation, significant ambiguity, non-canonical dialectal morphosyntactic constructions, and a wide variety of dialectal variants lacking established writing rules. Most NLP applications are seriously challenged by the underlying complexity of the Arabic language, which calls for ongoing research and innovation in this field [7].

Assessing sentiment in the Arabic language, particularly with its diverse colloquialisms like Iraqi Arabic, poses a significant challenge for traditional ML techniques. Frequently, ML methods lack sufficient efficiency due to the intricate characteristics of the language [8]. The complex structure and various dialects of the Arabic language create ambiguity, hindering machine comprehension. Moreover, the lack of annotated datasets for training amplifies the situation. They encountered difficulties due to the necessity of navigating the local context and sequential dependencies, which are crucial for comprehending Arabic text. Nonetheless, **Deep Learning (DL)** models, particularly transformer architectures exemplified by **Arabic Bidirectional Encoder Representations from Transformers (AraBERT)**, possess the capability to process the Arabic language

effectively. The embeddings generated by AraBERT effectively encapsulate context, enabling them to adeptly manage both local attributes and long-range dependencies inherent in Arabic [9].

This work presents a novel hybrid DL model that advances SA on the **Iraqi Arabic Dialect (IAD)** dataset by integrating AraBERT with **Convolutional Neural Networks (CNN)**, **Multi-Head Attention (MHA)**, and **Bidirectional Long Short-Term Memory (BiLSTM)**. The primary objective is to develop a model that can effectively classify feelings into positive, neutral, and negative categories, thereby achieving high performance metrics. The study conducted a systematic assessment of the impact of every architectural component by performing an ablation study in which the CNN, MHA, and BiLSTM are separately eliminated to evaluate their effects on general performance, as shown by the progressive reduction in metrics over Ablations 1 through 6. This work aims to provide practical insights on designing robust, multi-component DL architectures applicable to various challenging ANLP tasks, especially in the framework of dialectal SA, by clarifying the function of every component. The research paper is structured into several sections: the second section introduces related works, the third section presents dataset preparation, the fourth section presents the modeling detailed, the fifth part focuses on experiments and results, providing detailed analysis, and the final section presents the conclusions drawn from the study.

2 Related Work

With the application of DL approaches, recent developments in literature on ASA have been notable. This section reviews works published over the last five years (2020–2024), focusing on studies that align with our ablation methodology for evaluating the proposed framework on IQAD31K collected from Google Play Store and the IAD dataset proposed by [10]. Ablation studies, which methodically modify or eliminate model components to assess their impact on performance, are essential to fully understand the roles carried by architectural components and feature extractors in ASA.

Ghoul et al. [11] Explored three ML models to analyze sentiment in Arabic tweets. The first model uses both character-level and word-level features. The second leverages AraBERT along with Farasa preprocessing, and the third model combines these approaches into a **Voting Ensemble Classifier (VEC)** by using AraBERT embeddings. Findings demonstrate that the ensemble DL model enhances ASA.

Similarly, Alosaimi et al. [12] Introduced a hybrid DL structure that improves SA in Arabic text through the integration of AraBERT with LSTM networks. The authors proposed a model with a multi-stage architecture and AraBERT embeddings. The model was tested on four traditional Arabic datasets. The results indicate that the proposed AraBERT-LSTM model performed with an accuracy of more than 97%. The ablation study describes how each component integrates with the others, demonstrating that Arabs and LSTM layers significantly enhance the productivity of the process.

In a related study, Elhassan et al. [13] Explored DL techniques such as CNN, **Long Short-Term Memory (LSTM)**, and a combination of CNN-LSTM and word embeddings. They evaluated these models using two Arabic datasets: the **Hotel Arabic Reviews Dataset (HARD)** for hotel feedback and the **Large-Scale Arabic Book Reviews (LARB)** for book critiques. The CNN model achieved the highest accuracy rates of 94.69% on the HARD dataset when using the fastText word embeddings.

Meanwhile, Alqarni and Rahman [14] Explored how the COVID-19 pandemic has affected people's emotions in Saudi Arabia by analyzing Arabic tweets. The researchers employed CNN and BiLSTM algorithms, which achieved accuracies of 92.80% and 91.99%, respectively.

Furthermore, Mohamed et al. [7] Investigated ASA using DL techniques, specifically CNN, LSTM, a hybrid of CNN-LSTM, and BiLSTM. According to the study, the proposed model

outperforms previous models. The results show that LSTM and BiLSTM achieved an accurate value of 99.78%.

Additionally, Bello et al. [15] Presented a BERT-based model for SA of tweets, examining the effectiveness of transformer models. The suggested framework used the BERT model to classify tweets into sentiment categories. The research highlights the importance of deep contextual understanding in sentiment classification for social media content. The study methodology provides a robust framework for adapting BERT-based architectures to various languages and dialects, including Arabic.

Similarly, Matrane et al. [16] Proposed an approach based on AraBERT word embedding for Moroccan dialect SA. The study introduced a comparative analysis of ML algorithms and DL algorithms. According to the results, BiLSTM outperformed ML algorithms by 15% accuracy.

Additionally, Bdeir et al. [17] Presented a framework for Arabic tweets multi-label classification using word embedding techniques and DL algorithms. The study used a 160k Arabic tweets dataset from X and compared two DL methods, CNN, and **Recurrent Neural Networks (RNN)**. The results showed 90% accuracy scores and a Hamming loss of 0.02 for both networks.

Chouikhi et al. [18] Introduced an enhanced AraBERT tokenizer model designed for SA, explicitly targeting the difficulties arising from the intricate structure, many dialects, and limited resources of the Arabic language. The suggested methodology incorporates an Arabic BERT tokenizer, utilizing hyperparameter optimization and a random search technique to achieve optimal outcomes across various datasets. The investigation revealed that the BERT-based approach outperformed traditional ML and DL models, indicating the need for further endeavors to surmount these obstacles.

Moreover, He [19] Introduced a hybrid DL model for SA, integrating contextual embeddings from BERT, local feature extraction through CNN, and sequential pattern modeling via BiLSTM. The research demonstrates that this integrated architecture greatly enhances sentiment classification performance on benchmark datasets when compared to traditional and standalone DL models. Ablation experiments validate the distinct contribution of each component, underscoring the importance of integrating the individual contributions of each element. The designed principles provide a foundation for adapting the framework to accommodate linguistically diverse domains, such as Arabic dialects.

Additionally, Li et al. [20] Proposed a model built for evaluating the sentiment of reviews for the Chinese stock market. The framework has three main parts: BERT, for deep contextual embeddings; BiLSTM, to find sequential dependencies, and an attention mechanism to bring out features that are important for AS. Although the study focuses on Chinese text, its hybrid architecture provides insights for adapting SA models to other languages, including Arabic dialects.

Likewise, Farha et al. [21] Explored ML and DL approaches for ASA. The study evaluates **Naive Bayes (NB)**, **Support Vector Machine (SVM)**, CNN, and LSTM on benchmark datasets. The results show BiLSTM and CNN-LSTM are the most effective, with BiLSTM achieving F1-scores of 0.63, 0.72, and 0.89 on the SemEval, ASTD, and ArSAS datasets, respectively.

In addition, Elsamadony et al. [22] Examined ASA using Word2Vec word embedding technique and CNN-RNN model. The proposed model achieved an accuracy of 91.11%. The study showed that the XG-Boost classifier outperformed other methods on the LARB dataset.

Furthermore, Abdelminaam et al. [23] examined DL models (mLSTM, LSTM, and **Gated Recurrent Unit (GRU)**) and ML models (**Decision Tree (DT)**, **Logistic Regression (LR)**, **K-Nearest Neighbors (KNN)**, **Random Forest (RF)**, SVM, and NB) for ASA using three corpora. The study used **Term Frequency-Inverse Document Frequency (TF-IDF)** for ML and AraVec word embeddings for DL to handle feature extraction. Experimental results show mLSTM outperforms other models.

Moreover, Husain et al. [24] Presented “q8SentiLabeler,” a tool designed to improve SA for the Kuwaiti dialect. The proposed system aims to utilize a comprehensive dataset of over 16,600 posts, encompassing a wide range of topics and time periods. When tested, q8SentiLabeler showed the best performance, achieving an agreement rate of 93% and a Cohen’s Kappa coefficient of 0.87.

Along the same lines, Sabbah and Fasihuddin [25] Investigated different techniques like GloVe, Word2vec, and FastText with DL models such as BiLSTM and CNN for classifying sentiments in Arabic text. The results show that BiLSTM outperforms CNN.

Moreover, Matrane et al. [26] Introduced a stacked weighted-voting framework to annotate sentiment in Moroccan Arabic text automatically. The study trained a group of recurrent baseline algorithms and then aggregated their predictions on unlabeled targets. To strengthen representations, the model leverages AraBERT and DarijaBERT with dialect-specific vocabularies. Three fusion strategies are compared: the unweighted voting, metric-weighted voting, and a meta-learner. According to the study, the meta-learner achieved the highest agreement across six public datasets. The study further shows that dataset characteristics influence annotation quality.

Many studies have made comparative analyses to evaluate the effectiveness of various DL architectures and provide insight into the relative contribution of multiple components. For instance, Elhassan et al. [13] Compared DL techniques such as CNN, LSTM, and a hybrid CNN-LSTM model on two Arabic datasets, finding that the CNN model achieved the highest accuracy when using fastText as word embeddings, suggesting that the choice of embedding can significantly influence performance across architectures. Similarly, Mohamed et al. [7] Through a comparative study, examine the CNN, LSTM, CNN-LSTM, and BiLSTM models for ASA. The results show that LSTM and BiLSTM demonstrate the highest accuracy, highlighting the benefits of sequential modeling in capturing sentiment nuances in Arabic text. Additionally, Matrane et al. [16] Introduced a comparative study using AraBERT word embedding for Morocco dialect SA, showing that ML algorithms were outperformed by 15% in accuracy by BiLSTM. These comparative studies suggest that AraBERT and DL architecture improve ASA performance in Arabic datasets and dialects.

Collectively, the literature, while demonstrating progress, consistently reveals a persistent gap in architectural and dialectal performance in ASA. Existing models either over-rely on high-resource dialects or lack the necessary integrated complexity to robustly handle the extreme noise and morphological ambiguity found in under-resourced dialectal text, such as the IAD. This critical shortcoming, the absence of a single, thoroughly fine-tuned hybrid framework that synergistically leverages contextual embeddings, local feature filtering, and global dependency modeling, has precisely motivated our investigation. Our proposed work addresses this gap by introducing the novel AraBERT_CNN_MHA_BiLSTM architecture and conducting a systematic ablation study to precisely quantify how the integrated feature extraction mechanisms provide the required robustness for achieving superior performance in dialectal ASA.

3 Data Preparation

3.1 Datasets

The dataset used in this study comprises 31,324 reviews generated by Iraqi Arabic speakers, obtained from the Google Play Store. The reviews evolved from many different applications. These reviews were sourced from nine distinct applications, as detailed in Table 1, to ensure the dataset is suitable for SA.

As part of a comprehensive data cleaning process, questions, recommendations, and informational statements were screened out and categorized. These categories were covered using keyword detection and contextual analysis, with specific labels assigned to each (e.g., “Question,” “Suggestion,” and “Informative”). Additionally, the data preparation involves data cleaning to

Table 1. Applications and Review Counts for the Iraqi Arabic Dataset from Google Play Store

No.	Application Name	Reviews Number
1	AsiaCell	6,300
2	Dalil Iraq	1,168
3	First Iraqi Bank	753
4	IQ Passport	895
5	IQ Service	12,295
6	National Bank of Iraq	819
7	Rasheed Bank	3,650
8	Zain Cash	4,506
9	Taqaudi	938
	Total	31,324

Table 2. Inter-Annotator Agreement metrics

Metric (%)	Value
Agreement	98
Cohen's κ (quadratic)	96
Krippendorff's α (ordinal)	96

enhance the dataset's quality and reliability. This phase includes several sub-processes: removing diacritics, URLs, and filtering punctuation, hashtags, and stop words to eliminate noise; and spellchecking to correct linguistic errors. With entries labeled as positive, negative, or neutral depending on a method that considered linguistic nuances unique to the IAD, each record in the dataset underwent hand annotation by two native speakers to obtain high-quality sentiment classification. Furthermore, some words in **Modern Standard Arabic (MSA)** may have different meanings in Iraqi Arabic; therefore, this process examined dialect-specific expressions, colloquialisms, and sentences. The processing steps helped correct irregularities in the dataset, making it suitable for SA. Table 4 illustrate samples of the final dataset with sentiment labels. The cultural and linguistic context was integrated to ensure accurate sentiment labeling, resulting in a fully annotated dataset tailored for SA in this dialect. Statistical analysis of the dataset reveals a distribution of sentiments, providing a comprehensive representation of user sentiments, although the percentage of neutral reviews is relatively low, as demonstrated in Table 5.

The second dataset used in this study is the IAD from [10], which comprises 1,000 positive and 1,000 negative manually annotated comments taken from different Iraqi Facebook pages covering multiple fields, including home appliances, airways services, news page, restaurant, sports page, and a communications service.

3.2 Dataset Annotation

A sentiment polarity annotation scheme was implemented to construct the IQAD31K corpus. We engaged two Iraqi Arab native speakers to serve as annotators for comments extracted from Google Play Store Applications. To illustrate the nature of the annotated dataset, a sample of the dataset is provided in Table 2, which includes both the English translation and the sentiment label. Each comment was independently classified into one of three sentiment categories: Positive, Negative, or Neutral. This dual-annotator approach ensured a reliable gold standard for sentiment

Table 3. Confusion Matrix

	A2_negative	A2_neutral	A2_positive	Row total
A1_negative	60	1	0	61
A1_neutral	7	95	3	105
A1_positive	2	1	610	613
Column total	69	97	613	779

Table 4. Sample Iraqi Arabic Reviews from Google Play Store with Sentiment Classification

Review Translation	Review Text	Review Class
A helpful app, well done to the one who made it.	تطبيق مفيد عاشت ايد الي سواء	Positive
An excellent app, I recommend downloading it.	تطبيق ممتاز انصح بتحميله	Positive
An easy app with everything available on it.	بي موجود وكلشي سهل تطبيق	Positive
A terrible app, its services are really terrible and frustrating.	تطبيق كلش تعبان وخدماته كلش تشلع الكلب	Negative
The worst app in history.	أنكس تطبيق بالتاريخ	Negative
A very failed and useless app.	تطبيق فاشل جدا وغير مفيد	Negative
Under testing.	تحت التجربة	Neutral
Let's try it and see.	نجرب ونشوف	Neutral
Let's check it out, hopefully it's worth it.	خل نشوفه بالله يسوة	Neutral

Table 5. Sentiment Class Distribution of Iraqi Arabic Reviews from Google Play Store

Review Class	Review Number	Percentage (%)
Positive	18,032	57.57
Negative	12,756	40.72
Neutral	536	1.71
Total	31,324	

classification, making it possible to calculate inter-annotator agreement. Two annotators labeled 779 randomly selected IQAD31K reviews as negative, neutral, or positive in sentiment. Agreement was 98%, quadratic-weighted Cohen's $\kappa=0.96$, and Krippendorff's α (ordinal) was 0.96 as demonstrated in Tables 2 and 3.

To guarantee that the annotation steps were as accurate and reliable as possible, a structured training and quality control process was employed. A theoretical overview of SA and various classification types was provided to ensure that all participants understood the context and their role. After that, the annotators worked together to annotate five sample comments, looking for details to make sure everyone understood the rules for classification and to eliminate any ambiguity.

Subsequently, independently, both annotators labeled an initial set of comments. Then, a review session was conducted to determine if they agreed and to address any disagreements. Disagreements ($n=14$) occurred mainly at the neutral boundary. After establishing inter-rater reliability, the annotators independently classified the corpus. For experiments, we resolved all disagreements by expert adjudication and used the adjudicated labels as the gold standard. This protocol made sure that the IQAD31K corpus had a high-quality, fair, and trustworthy gold standard [27].

As shown in Table 5, the dataset is distributed across sentiment categories as follows: The positive instances represent 18,032 reviews (57.57%), followed by 12,756 negative reviews (40.72%), while neutral reviews are markedly underrepresented, accounting for 536 (1.71%). This imbalance

Table 6. Comparison of Iraqi Arabic review Before and After the Preprocessing Phases

Meaning in English	Before preprocessing	After preprocessing
I downloaded it in support of my country good luck	نزله دعما لبلدي بالتوفيق العراقي يظل راقى 🇮🇶	نزله دعما لبلدي بالتوفيق العراقي يظل راقى
Iraqi remains classy		
Best Iraqi Application	أفضل تطبيق عراقي 🇮🇶 ❤️	أفضل تطبيق عراقي

in representation is inherent to the data source, which is dominated by explicit positive and negative opinions.

3.3 Preprocessing steps

To enhance the quality of the collected Iraqi Arabic text data and ensure optimal model performance, a multi-stage, detailed preprocessing pipeline was implemented. Initially, using Panda's library, the raw dataset was uploaded and read. The preprocessing strategies were directly designed to address the linguistic challenges associated with the Arabic language, especially those related to the Iraqi dialect. The initial stage included the Unicode normalization standard. It was necessary to standardize the character structure and maintain consistent text representation. Subsequently, diacritics, and URLs were removed, which often introduces unnecessary noise.

The standardization of Arabic orthographic variants was the next step in the text normalization process. Using the CAMEL Tool [28] library the key transformations included the normalization of hamza forms (إ, ا, آ) into a simple alef (ا) and the conversion of (ة) into (ه). Additionally, to address frequently encountered variations a specific word normalization was applied, such as converting words ending in (ى) to (ي) and standardizing the word "شيء" (which means "thing") into "شي". Following normalization, punctuation marks, special characters, and non-Arabic characters were eliminated to ensure that the resulting corpus exclusively contained Arabic text to enhance model training.

A comprehensive collection of stopwords was assembled, combining traditional Arabic stopwords with those specific to the Iraqi dialect, such as "شلون" meaning "how", "شنو" meaning "what", and "هسه" meaning "now". To retain only the most semantically relevant tokens, stopword removal was performed, thereby improving feature quality and reducing data sparsity, as illustrated in Table 6. This cautious preprocessing ensures that the dataset is linguistically normalized, synthetically clean, and meaningful, facilitating the development of a stronger and more accurate sentiment classification model.

3.4 Train validation and test setup

The standard train/validation/test partitioning strategy was employed in this study to train the proposed model in various modes, ensuring that its performance is robust across all datasets. Stratification technique was used to retrieve the split made by train_test_split function of scikit-learn in such a way that the sentiment distribution of the entire dataset was kept intact among the train, test, and validation subsets [29].

The process starts with the initially available dataset being divided into training and an evaluation dataset. 80% of the dataset was used as a training set, whereas the 20% was further divided into two parts, 10% for the validation set and 10% for the test set, using the stratification splitting method. This approach ensures to keep the positive, negative, and neutral sentiment classes distribution at almost the same proportion across all subsets, thus reducing class imbalance bias.

The training set was used to adjust the proposed model parameters, while the validation set was used to optimize hyperparameters. We perform early stopping to prevent overfitting by monitoring

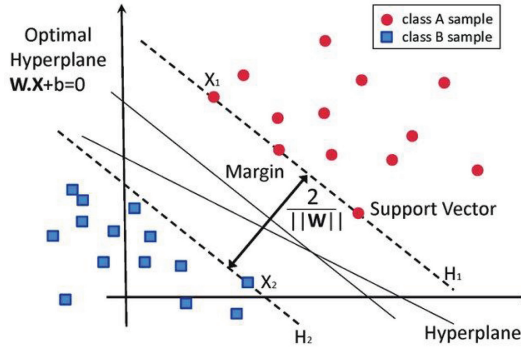


Fig. 1. Classification of data by SVM [32].

validation loss, accuracy, precision, recall, and F1-score. The test set is used to evaluate the model's performance on unseen data. The 80/10/10 partition in addition to the stratification helps to train and validate the model effectively, and also evaluates its performance on the new data that has a complete composition of the dataset's sentiment distribution [30].

4 Modeling

4.1 Machine Learning Algorithms

4.1.1 Random Forest. RF algorithm is a supervised learning approach, that is, capable of handling both classification and regression problems by collecting the results of many individual decisions. Within the model, each tree makes a prediction depending on the input features, and the overall decision is determined by the majority voting on all trees. This method depends on a mathematical model that merges several decisions to merge trees into integrated trees. The formula depends on how many trees are included in the forest [31]. Below is a mathematical expression that defines a RF classifier:

$$n_{ij} = W_i C_j - W_{\text{left}(j)} C_{\text{left}(j)} - W_{\text{right}(j)} C_{\text{right}(j)} \quad (1)$$

where:

n_{ij} denotes the significance of node j .

$w_{sub}(j)$ refers to the weighted number of samples that arrive at node j .

$C_{sub}(j)$ represents the impurity value of node j .

RF was implemented using the scikit-learn library, with the `Random_State = 42` parameter specified to ensure reproducibility of the results. No further hyperparameters were explicitly defined, which involves the adoption of standard values. This ensemble-based configuration was applied to AraBERT embeddings to improve classification robustness by balancing variance and bias in the SA task.

4.1.2 Support Vector Machine. The SVM is a supervised learning algorithm used to identify optimal decision boundaries for separating distinct data clusters. Specifically, SVM seeks to determine the most effective hyperplane by maximizing the margin between different data groups, thus improving classification accuracy as shown in Figure 1. The hyperplane serves as an interclass separator, defining the boundary between classes within the feature space. Mathematical composition is represented in Equation (2) to obtain the best hyperplane, establishing a comprehensive foundation for achieving optimal differentiation among classes. This approach ensures robust classification by placing the hyperplane to obtain the most significant

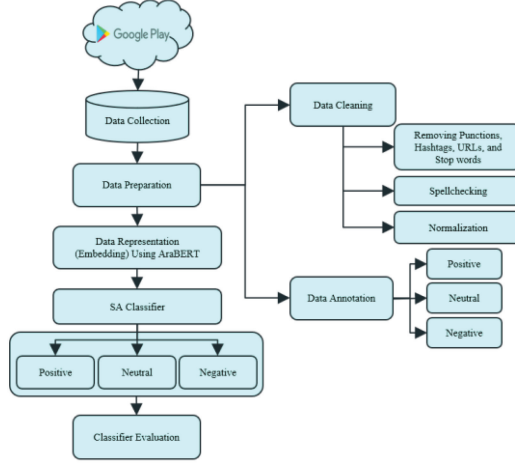


Fig. 2. The proposed model for Iraqi Arabic SA.

possible distance from the nearest data points in each class, thus minimizing the risk of error classification [32].

$$w \cdot x + b = 0, \quad (2)$$

where w is the weight vector (normal to the hyperplane), x is the input feature vector, and b is the bias (intercept). The distance from the sample point to the hyperplane is referred to as the margin, which can be expressed in Equation (3).

$$\text{Margin} = \frac{1}{\|w\|}. \quad (3)$$

The SVM classifier has been implemented using Scikit-Learn by adjusting the probability to True. This setting enables the calculation of the probability estimates, which are required to calculate the log-loss matrix during the model evaluation. Other hyperparameters were used as defaults for the SVC class, which typically includes the use of a **Radial Basis Function (RBF)** kernel and a regularization parameter $C = 1.0$. This setup ensures that SVM can effectively handle the non-linear isolation of embedded features by providing potential outputs for performance evaluation.

4.2 Proposed Hybrid Model Architecture

The proposed framework illustrates a pipeline for SA on user-generated content for the IAD collected from the Google Play Store, as demonstrated in Figure 2. The process begins with data collection, followed by the data preparation phase. The AraBERT model was used to obtain contextualized embeddings that capture the semantic features of Iraqi Arabic text. The model fine-tunes the pretrained encoder together with the added layers, rather than using the encoder only to produce fixed embeddings. During training, errors propagate through the entire network, updating all weights, which helps the representations adapt to the task and often improves accuracy compared to frozen features. Additionally, DL algorithms were deployed for feature extraction and modeling. The model leverages a combination of CNN for feature extraction, MHA for capturing global dependencies, and BiLSTM networks for sequential feature learning. The combination of these architectures enables the model to learn both local and global contextual information in Iraqi Arabic text with effectiveness. Ultimately, the stage of classifier evaluation assesses the model's performance across the sentiment categories. This evaluation ensures that the trained model can

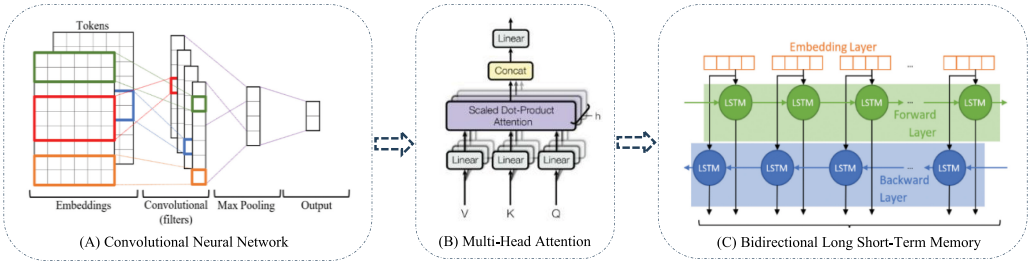


Fig. 3. Unified architecture of the proposed model for Iraqi ASA.

typically predict user sentiment and generalizes successfully on unseen data, therefore providing significant information for academics in the field of **Arabic natural language processing (ANLP)** and application developers.

Figure 3 represents the unified structure of the proposed AraBERT-CNN-MH-BiLSTM on the SA with the Iraqi Arabic. The architecture starts with the embeddings obtained through AraBERT, and then these are processed by CNN to obtain local lexical features. These are then passed to a **multi-headed (MH)** attention module that helps to contextually weight representations. Next, a BiLSTM layer is used to model the sequential dependencies. Lastly, the enhanced feature representations are passed through fully connected layers for final sentiment classification outputs.

The transformer-based model AraBERT was introduced by [9]. It is pre-trained for Arabic language recognition using both MSA and dialectal Arabic datasets, which addresses the unique challenges of ANLP. Historically, Arabic has presented challenges for tasks such as SA, NER, and QA due to its morphological richness and syntactic complexity, as well as its lexical sparsity and limited resources. AraBERT is based on the BERT architecture and includes 12 encoder blocks, 768 hidden dimensions, 12 attention heads, and a maximum sequence length of 512. This configuration has a total of approximately 110 million parameters [33].

The pre-training process utilizes a vast collection of Arabic material, comprising millions of sentences gathered from various sources, including local news and language databases. To address Arabic morphology, the model utilizes a specialized segmentation tool that breaks words into their basic stems, prefixes, and suffixes. This approach simplifies the tokenization process and reduces redundancy, making it more efficient and manageable. By constructing a large and representative vocabulary, the model is better equipped to capture the rich linguistic diversity and nuances present in Arabic text.

During the pre-training stage, AraBERT was trained on large Arabic text, optimizing BERT's standard objectives: **Masked Language Modeling (MLM)**, with whole-word masking and **Next Sentence Prediction (NSP)**. This approach aims to enhance AraBERT's ability to handle ANLP tasks.

AraBERT was tested across multiple standard benchmark datasets for Arabic NLP, and AraBERT has outperformed both multilingual BERT and previous state-of-the-art models by being more efficient in SA, NER, and QA tasks. During the SA task, the model recorded 96.2% accuracy and recognized a named entity with an F1-score of 84.2 [9].

Word embedding methods have been shown to be instrumental in the process of ASA, with a record of outperforming other baseline conventional methods at the data representation step [13]. The representations made with these techniques correspond to word vectors, which, in turn, allows the neural networks to understand the meanings of words semantically. Moreover, DL based word embeddings reflect semantic and syntactic contexts, which are helpful in many tasks, including ASA [34]. As a result, the present paper utilizes the AraBERT word embedding model for the data

representation phase of the proposed framework, particularly as it can extract specific linguistic features that are significant for accurate sentiment classification in Arabic [35].

The CNN component employed in the proposed architecture is crucial because it helps extract local features from embeddings contextualized by AraBERT. When AraBERT completes the input text and produces the embeddings, the CNN layer configured as 1D convolution operates on these embeddings to achieve n-gram-like patterns, as shown in Figure 3. CNN takes the embeddings, and a convolution is applied that preserves the sequence length with the help of padding. Through the operation of this type, one can see local features that focus on small, contiguous regions of the sequence, which are later searched for the syntactic and semantic patterns resulting in meaningful representations, such as phrases, word combinations, correlated to sentiment [36].

The ReLU activation function enables the model to incorporate non-linear aspects, which are crucial for effectively capturing intricate feature interactions. After the convolutional process is completed, the establishment of the LayerNorm technique supports stable training progression by not only normalizing the output but also by treating the internal covariate shift problem. The feature map, that is, produced still has local contextual details that have not been lost, which are crucial in the SA domain. In SA, the feature map, that is, local in nature provides a local context, that is, augmented by the global one of the later layers; thus, subsequently, the model is improved in terms of the possible identification of sentiments that would be found in the morphologically complex IAD [37].

Once CNN generates local features, Multi-Head Self-Attention is used to identify long-range dependencies in feature sequences. MHA computes attention scores across multiple subspaces to capture relationships across all tokens, regardless of their position. As demonstrated in (B) Figure 3, this allows the model to simultaneously emphasize several sequence elements and find contextual ties like distant words influencing sentiment. MHA enhances SA, where semantic and discourse-level relationships extend sentences [38]. Moreover, BiLSTM in the proposed model is employed for controlling sequential dependency detection. In the MHA phase, which captures only long-range dependencies, the BiLSTM passes through the sequence to detect temporal relationships. This bidirectional method, as shown in (C) Figure 3, improves token accuracy and help to understand each word by analyzing context from other words, which is crucial for ASA. The BiLSTM gates solve the vanishing gradient problem, allowing them to withstand long-term pattern information relevance differences [39].

4.3 Ablation Study

The full version of the proposed framework is a composite of AraBERT, connected with a CNN, MHA, and BiLSTM, to perform SA in the IAD. The first step that AraBERT follows is creating embeddings from the given text, which will capture in-depth language characteristics of Arabic. The CNN, in turn, searches for local n-gram-like patterns that are key to recognizing syntactic and semantic structures necessary for the task of sentiment detection. The MHA mechanism is next in line, which identifies long-distance dependencies and enables the model to comprehend the context between tokens that are far apart in the sequence. Finally, the BiLSTM is the component that detects the order of a sequence in both directions, which results in understanding all temporal dynamics in the sentence. The unified architecture here benefits from each constituent, including local feature extraction, global context awareness, and sequential modeling, which collectively address the morphological and syntactic complexities of Arabic and are the primary factors driving robust performance for the model.

The training process for the proposed model, aimed at analyzing sentiment in the IAD, is carefully crafted to enhance performance while keeping computational limits in mind. The model operates with a batch size of 32, striking a balance between memory efficiency and gradient stability

throughout the training, validation, and test phases. The dataset divides into 80%, 10%, and 10%, thus these phases are arranged accordingly. We leverage the adaptive learning properties of the AdamW optimizer with a learning rate of $2e-5$ to update the model parameters, so efficiently managing the sparse gradients typical in transformer-based models such as AraBERT. The training process was set up to run up to a maximum of 100 epochs, providing multiple opportunities for the model to converge. To ensure that the process is efficient, an early stopping callback is implemented to trigger after 15 epochs without any improvement in the validation loss preventing overfitting. The gradient accumulation technique is employed with a step size of 2, thereby simulating a larger batch size by accumulating gradients for two iterations before the model weights are updated. This technique is especially helpful in optimizing the use of memory for GPU resources. Moreover, mixed precision training is being employed using torch.amp, which allows computations to be carried out in 16-bit and 32-bit floating-point formats. This not only accelerates training but also eliminates memory usage without compromising the model's performance. Additionally, a ReduceLROnPlateau scheduler halves the learning rate if the training loss is not improving for 2 epochs offering appropriate convergence. Altogether, this holistic training approach is designed to improve the performance of the model for the difficult task of ASA.

The proposed model takes a thorough approach to evaluation by using a variety of evaluation metrics like Accuracy, Precision, Recall, and F1-score to measure its performance in ASA, specifically on the IAD dataset. These metrics are calculated in the evaluation function, which analyzes the model's predictions from both the validation and test sets. Accuracy represents the percentage of properly classified cases and calculated as the number of accurate predictions divided by the total number of samples. Though it can provide an overall view of the performance of the model, occasionally it can be misleading, particularly in cases of unbalanced datasets. Focusing on the ability of the model to lower false positives across the three sentiment classes is a key aspect of this evaluation. Precision, computed as a weighted average, measures the ratio of correct positive predictions with a focus on the model being able to limit false positives among the three sentiment categories: positive, neutral, and negative. Recall shows the number of real positives that are correctly identified, consequently, illustrating the need to discover all positive cases in the model. Using an average of precision and recall, the F1-score addresses false negatives and false positives equally, accordingly, benefiting unbalanced data. Together, these metrics give a clear picture of the model's performance, covering both overall accuracy and how well it does with different classes, especially in low-resources ASA.

5 Experiments

5.1 Implementation

The experiments for the proposed model were conducted within a solid computational framework that utilized PyTorch, Hugging Face Transformers, and Kaggle. This setup ensured efficient implementation and facilitated the deployment of our ASA model on the IAD dataset. This setup provided an efficient framework for preprocessing, data loading, model training, and evaluation with consistent generation of results. Table 7 presents the essential hyperparameters that characterize the proposed model and training environment, as specified for complete reproducibility.

5.2 Experimental Setup

A dataset extracted from the Google Play Store was used in conjunction with the IAD dataset. This includes processed sentences labeled with sentiment categories: positive, neutral, and negative. To support the evaluation of the appropriate model, the dataset was split into training, validation, and test sets. The proposed model combines AraBERT with a CNN, an MHA, and a BiLSTM network.

Table 7. Proposed model Hyperparameter

Component	Hyperparameter	Value
Architectural Core	AraBERT Model	AraBERT v1 weights
	CNN Filter Size	3
	CNN Output Channels	1,024
	MHA Num Heads	8
	BiLSTM Hidden Units	128
	BiLSTM Layers	2
Training Process	Learning Rate	2.00E-05
	Batch Size	64
	Fine-tuning Strategy	Full Fine-tuning
	Early Stopping Patience	15
Data Handling	Max Sequence Length	128
	Random State/Seed	42
Regularization	Dropout	0.5

It concludes with a fully connected layer for sentiment classification. To stabilize training and prevent overfitting, the study employed a normalization layer and dropout techniques. Early stopping and learning rate modification techniques were used to improve model performance and efficiency. Evaluation metrics, including accuracy, precision, recall, and F1-score, were used to thoroughly evaluate the model's performance and determine its suitability for ASA.

In the first step of the ablation process, the three parts of the DL system are discarded, and the model is employed as a traditional ML method through the use of SVM. AraBERT was used as a word embedding to produce feature vectors as an input to the SVM. This process aims to reveal the role played by AraBERT's contextual embeddings through the elimination of the DL layers. However, the linear classifier, which is the SVM, strictly depends on the static embeddings and thus it lacks the capacity to handle difficult co-occurrences or the modelling of sequential context as in the whole model.

In the second phase of the ablation study, the DL model was replaced by a RF classifier. The implementation method is the same as in Ablation 1, where AraBERT embeddings are used as input features for SA on the Iraqi Arabic dataset. The RF can't recognize sequential relationships and distant correlations, although it utilizes dynamic information from the context provided by AraBERT. The goal of this test is to demonstrate the effectiveness of AraBERT embeddings in a non-sequential, tree-based approach and to highlight that traditional ML methods struggle to analyze changing Arabic sentiment patterns.

In the third step, the AraBERT was used as baseline for SA in Iraqi Arabic text, allowing us to study the algorithm outcomes.

In the fourth step of the ablation, the CNN component, in addition to AraBERT, was used for SA of the IAD dataset. In this case, the layers MHA and BiLSTM are entirely removed, allowing us to draw the conclusion that the local feature extraction stage is the one that matters most. As for AraBERT, its embeddings are context-based and are subsequently detected by a CNN to identify linguistic terms, which are sequences of words that represent a sentiment.

In the fifth ablation step, the CNN was dropped, and AraBERT, MH, and BiLSTM were employed for SA. AraBERT induces embeddings specific to the context, and MHA utilizes them directly to capture long-range dependencies among the tokens of a sequence, which, as a result, allows the model to understand the relationships between far-off tokens. After that, the BiLSTM layer

Table 8. Ablation study results in percentage points

	Approach					IQAD31K dataset				IAD dataset			
	SVM	RF	CNN	BiLSTM	MH	A	P	R	F1	A	P	R	F1
AraBERT_SVM	✓					92.95	93.05	92.95	92.85	89.50	89.82	89.50	89.48
AraBERT_RF		✓				91.22	91.47	91.22	91.02	88.00	88.56	88.00	87.96
AraBERT (Baseline)						92.48	92.48	92.48	92.46	93.00	93.02	93.00	93.00
AraBERT_CNN			✓			95.56	95.56	95.56	95.55	92.50	92.54	92.50	92.50
AraBERT_MHA_BiLSTM				✓	✓	95.50	95.52	95.50	95.46	93.00	93.16	93.00	92.99
AraBERT_MHA_CNN			✓		✓	95.56	95.57	95.56	95.55	94.00	94.16	94.00	93.99
AraBERT_CNN_BiLSTM			✓	✓		95.60	95.58	95.60	95.58	93.50	93.61	93.50	93.50
AraBERT_CNN_MH_BiLSTM			✓	✓	✓	95.63	95.67	95.63	95.63	94.50	94.50	94.50	94.50

The bold entries highlight the comparative strength of each hybrid configuration. CNN consistently provides the largest performance, while adding BiLSTM and Multi-Head Attention refine results. The best-performing model is **AraBERT_CNN_MH_BiLSTM**, underscoring the importance of combining local, sequential, and global contextual features for Iraqi Arabic sentiment analysis.

creates word sequences as output to model the sequential dependencies in either direction, thereby capturing the temporal dynamics of the sentence. This part also scrutinizes the cross-effectiveness of MHA and BiLSTM in tackling Arabic's difficulties and illustrates the lack of patterns extracted by CNN.

In the sixth step, the model is set up as AraBERT_MH_CNN, which eliminates the BiLSTM part while still utilizing AraBERT, MH, and CNN for SA. AraBERT produces the context representation and then passes it to CNN for the extraction of local features that are related to the sentiment. After that, the MHA layer is used to trace topics with long distances to establish the relationship of the tokens in the sequence. In the absence of BiLSTM, the model lacks the ability to represent sequence order explicitly, yet it retains CNN's local feature extraction and MH's global context awareness. This ablation allows us to investigate the combined effects of local and global feature extraction in the absence of sequential modeling.

In the last phase of the ablation step, only the MHA component is not implemented, while still being consistent with AraBERT, CNN, and BiLSTM for the SA of the IAD dataset. In the first step, AraBERT produces contextual embeddings. CNN then consumes these embeddings to get critical local features. Finally, BiLSTM effectively captures the temporal context of the sequence from both directions due to its dual nature. This ablation study investigates the joint contribution of local feature extraction and sequential modeling, highlighting the role of MHA in recognizing global relationships within sequences.

5.3 Results

5.3.1 Full Model: AraBERT_CNN_MH_BiLSTM. The comprehensive model, incorporating CNN, MH, and BiLSTM layers, achieves the highest performance across all metrics and datasets. It achieves 95.63% accuracy, 95.67% precision, 95.63% recall, and 95.63% F1-score on IQAD31K, and 94.50% across all metrics on the IAD dataset, as demonstrated in Table 8. These results indicate a well-balanced and normalized architecture, that is, capable of maintaining high performance across datasets. The superiority of this model over the other ablation experiments is due to the effective coordination between contextual embedding (AraBERT), local feature extraction (CNN), attention mechanisms (MHA), and sequential modeling (BiLSTM), each contributing significantly to the overall performance.

5.3.2 Ablation 1: AraBERT_SVM. The SVM algorithm demonstrated a low performance among all models tested. The model reaches only 92.95%, 93.05%, 92.95%, and 92.85% on IQAD31K and 89.50%, 89.82%, 89.50%, and 89.48% on the IAD dataset for accuracy, precision, recall, and F1-score, respectively. While this improves the SVM's performance compared to RF, its performance is still inferior to that of any computationally intensive learning model. This demonstrates the SVM's limited ability to handle sequential and relevant data, which contributes to its inferior results, thereby strengthening the need for modern NLP architectures in complex SA functions.

5.3.3 Ablation 2: AraBERT_RF. Changing DL components with a classical ML model, RF, leads to a significant decline in performance. The evaluation metrics scores fall to 91.22%, 91.47%, 91.22%, and 91.02% on IQAD31K and 88.00%, 88.56%, 88.00%, and 87.96% on the IAD dataset for accuracy, precision, recall, and F1-score, respectively. These results outline the range of traditional ML methods in capturing complex syntactic and semantic features present in the text. Unlike DL approaches, RFs cannot process relevant information, which is crucial for nuanced SA and objective evaluation.

5.3.4 Ablation 3: AraBERT (Baseline). In our ablation study, we used AraBERT as a DL baseline model following the evaluation of ML classifiers. The baseline model achieved an F1-score of 92.46% on IQAD31K and 93.00% on the IAD dataset, as shown in Table 8. This shows a significant performance increase compared to the ML classifier. This improvement ensures the role of the transformer base architecture and its contextual pre-training in understanding the contextual and morphological nuances of dialectal Arabic.

5.3.5 Ablation 4: AraBERT_CNN. In this configuration, both MHA and BiLSTM were excluded, maintaining only AraBERT and CNN. Despite the simplified architecture, the model performs well, achieving 95.56%, 95.56%, 95.56%, and 95.55% accuracy on IQAD31K. However, it decreases to 92.50%, 95.52%, 92.50%, and 92.50% accuracy on the IAD dataset for precision, recall, and F1-score, respectively. This indicates that although CNN effectively captures local characteristics, the absence of MHA and BiLSTM reduces the model's ability to comprehend deep meaning and sequential relationships, which are crucial for adapting to diverse or less familiar data contexts.

5.3.6 Ablation 5: AraBERT_MH_BiLSTM. In Ablation 4, the CNN component is excluded from the complete model, resulting in a slight decrease in performance. The model still performs well, achieving 95.50%, 95.52%, 95.50%, and 95.46% accuracy on IQAD31K, and 93.00%, 93.16%, 93.00%, and 92.99% on the IAD dataset for precision, recall, and F1-score, respectively. These results suggest that CNN contributes mainly to the extraction of local and position-invariant features. Although the remaining components (MHA and BiLSTM) may compensate to some extent, this demonstrates that CNN is valuable to improve generalization, especially when dealing with varied sentence structures or specific language patterns that commonly appear in datasets beyond the training set.

5.3.7 Ablation 6: AraBERT_MH_CNN. This version removes the BiLSTM layer, maintaining the AraBERT, MHA, and CNN components. The model maintains a high performance of 95.56%, 95.57%, 95.56%, and 95.55% on IQAD31K, but the performance on the IAD dataset decreases to 94.00%, 94.16%, 94.00%, and 93.99% for accuracy, precision, recall, and F1-score, respectively. These findings indicate that while BiLSTM plays an essential role in handling sequential dependence and temporal patterns. Therefore, BiLSTM enhances the model's strength, particularly in unseen data scenarios.

5.3.8 Ablation 7: AraBERT_CNN_BiLSTM. In Ablation 6, the MHA layer is removed from the whole model. Although the architecture still performs competitively, there is a slight but consistent decline in performance, especially on the external dataset, with 93.50%, 93.61%, 93.50%, and 93.50%

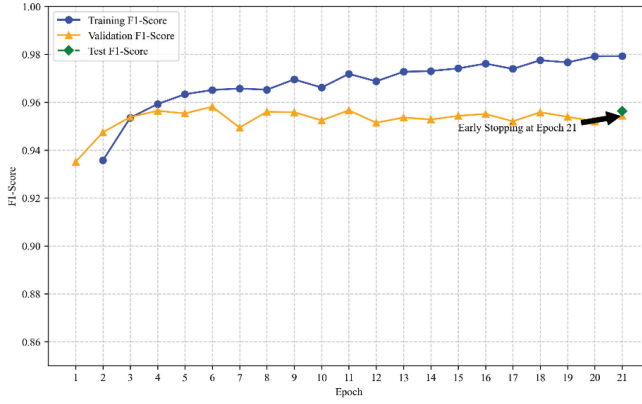


Fig. 4. Training, validation, and test f1-Score over epochs for the full model over IQAD31K dataset.

Table 9. Effect of Removing Each Layer in percentage points

	IQAD31K dataset				IAD dataset			
	A	P	R	F1	A	P	R	F1
Full Model - CNN	0.13	0.15	0.13	0.17	1.50	1.34	1.50	1.51
Full Model - BiLSTM	0.07	0.10	0.07	0.08	0.50	0.34	0.50	0.51
Full Model - MH	0.03	0.09	0.03	0.05	1.00	0.89	1.00	1.00

on the IAD dataset, and 95.60%, 95.58%, 95.60%, and 95.58% on IQAD31K for accuracy, precision, recall, and F1-score, respectively. This indicates that MHA contributes to improved generalization by enhancing the model's ability to focus on relevant contextual tokens. Its absence limits the model's ability to capture complex dependencies, especially in data that varies from the training distribution.

5.4 Discussion

The results of the ablation study, as illustrated in Table 8, provide significant insights into the individual and collective contributions of each component within the proposed model. Its purpose was to dissect architecture and understand how each layer affects the general performance, particularly when utilized in the complex and less standardized IAD. In particular, the whole model achieves advanced performance on both datasets. The robust accuracy of 95.63% on our customized dataset and 94.50% on the IAD dataset confirms the value of combining contextual embeddings, local feature extraction, global dependency modeling, and sequential learning. The stability and convergence behavior of the full model during the optimization phase on the IQAD31K dataset are further illustrated by the F1-Score progression across training, validation, and test sets over epochs, as depicted in Figure 4.

These results confirm that each architectural element plays a supplementary role in addressing the challenges that come with ASA. The relative impact of each component on overall performance is summarized in Table 9, quantifying the percentage point drop in metrics upon removal. When evaluating the effect of eliminating every component, CNN emerged as the most impactful. Excluding the CNN in Ablation 4 caused a noticeable drop in performance, especially on the IAD dataset, where accuracy declined to 93.00%. This suggests that the CNN's ability to capture nearby syntactic patterns and sentiment-indicative n-grams is crucial for dialectal interpretation, in which sentiment can hinge on brief, casual, and idiomatic terms.

Table 10. Execution Time for Each Ablation

No.	Ablation phases	Dataset	Exec. Time (s)
1	ML - SVM	IQAD31K	514.5
		IAD dataset	1.9
2	ML - RF	IQAD31K	86.1
		IAD dataset	3
3	AraBERT_CNN	IQAD31K	942.0
		IAD dataset	150.6
4	AraBERT_MHA_BiLSTM	IQAD31K	3,350.1
		IAD dataset	233.4
5	AraBERT_MHA_CNN	IQAD31K	3,053.1
		IAD dataset	155.2
6	AraBERT_CNN_BiLSTM	IQAD31K	3,672.7
		IAD dataset	389.8
7	AraBERT_CNN_MH_BiLSTM	IQAD31K	6,383.6
		IAD dataset	752.5

The absence of MHA in Ablation 6 resulted in a decline in performance, although less noticeable, as the accuracy of the IAD dataset declined to 93.50%. This indicates that long-range dependencies, such as clauses that stretch across many tokens and fragmented emotional cues, are heavily influential but non-negligible. The removal of BiLSTM in Ablation 5 also resulted in a slight accuracy decrease that implying the usefulness of BiLSTM to retain the sequential context of text where words may be ordered colloquially.

In contrast, substituting DL models with classical ML algorithms, RF and SVM, in Ablations 1 and 2 resulted in much poorer performance, especially on the IAD dataset. This underscores the limited capability of traditional models towards encoding the diverse morphosyntactic features embedded in Iraqi Arabic, as these features are substantially affected by dialectal irregularities and contextual variability. Notably, components such as AraBERT and the MHA module introduce significant computational overhead due to the complexity of transformer-based architectures and attention mechanisms; their exclusion leads to a marked decline in model performance. This emphasizes that even with the additional computational cost, there is a need to keep these components in construction. Future research can, consequently, explore more efficient alternatives or employ strategies like knowledge distillation to reduce inference time while preserving accuracy.

By analyzing the execution time presented in Table 10 for the ablation study, we can offer insightful information on the computational efficiency of several models' configurations for SA in two datasets. The full model has the longest execution time, requiring 643.6 seconds on the IQAD31K dataset and 752.5 seconds on the IAD dataset.

The full model's high accuracy comes with a significant computational overhead, as shown by its long execution times. For practical, low-latency deployment, the ablation study provides a solution, where lighter intermediate models (e.g., the AraBERT_CNN variant) offer competitive performance-to-cost trade-off. Future work will focus on optimizing the full architecture's inference speed using techniques like knowledge distillation [40].

This extended period is a sign of computational demands associated with a combination of transformer-based embedding, conventional layers for local feature extraction, attention mechanisms to capture long-distance dependence, and sequential modeling layers, which all contribute to its high performance.

Table 11. Comparison of the proposed model with baseline models on the IAD dataset

Models	Accuracy	Precision	Recall	F1-score
Proposed Framework	94.50%	94.50%	94.50%	94.50%
Alnawas and Arici [10]	–	82%	79%	78%
Anwar and Nada [41]	92.5%	89.6%	92.079%	92.537%

Conversely, the ablation configurations reveal a notable decrease in execution times when specific components are excluded or substituted with simpler methods. The first two ablation phases, which employ traditional ML approaches, exhibit the shortest execution times, with 514.5 seconds and 86.1 seconds on IQAD31K, and 1.9 seconds and 3 seconds on IAD, respectively. These reduced times reflect the lower computational complexity of traditional methods, which, however, come at the expense of performance. The subsequent ablation phases, which retain subsets of DL components, show intermediate execution times ranging from 942.0 to 3672.7 seconds on IQAD31K and 150.6 to 389.8 seconds on IAD datasets, highlighting a balance between computational cost and model effectiveness. For example, the third ablation phase, which focuses just on local feature extraction, requiring 942.0 seconds on IQAD31K, results in an F1-score of 92.50% on IAD. These results demonstrate a natural trade-off between computational efficiency and model performance in SA tasks. The results indicate that although detailed DL models show better performance, their computational overhead should be carefully considered.

The ablation investigation reveals that, in general, numerous components of the model depend on each other in fundamental ways; performance is not driven by a single module, but by the combination of several, including AraBERT, CNN, MHA, and BiLSTM. This compound structure is especially effective for processing the Iraqi Arab dialect, which presents unique challenges in terms of token meaning, the order of colloquial words and expressions, and the recognition of flexible patterns that require a deep contextual understanding.

We further interpret our findings by comparing the performance of the suggested approaches with previous baselines and state-of-the-art systems. The performance of the proposed framework on Iraqi Arabic text reveals a marked improvement over the two baseline models, as shown in Table 11, which [10] reported an accuracy of 82%, with a precision of 79%, a recall of 79%, and an F1-score of 78% on IAD dataset. In the second baseline by [41] the results achieved for the IAD dataset were an accuracy of 92.5%, a precision of 89.6%, a recall of 92.079%, and an F1-score of 92.537%. On the other hand, our proposed framework outperformed both baseline algorithms, achieving an accuracy of 94.50%, a precision of 94.50%, a recall of 94.50%, and an F1-score of 94.50% compared to the baseline models. This outcome demonstrates how the proposed hybrid DL approach compares to baseline models, consequently, advancing ANLP.

The performance of the proposed model is superior for the IAD dataset and surpasses the previous work reported by the authors in [10] and [41]. Compared to Anwar and Nada, our model achieves absolute merits of between +1.96 (in terms of F1-score) and +4.90 percentage points in Precision. They are increasing for all metrics by more than two absolute points for accuracy, recall, and F1-scores, while precision increases by a remarkable 5.47%. When compared to Alnawas and Arici, the results are more positive. The absolute performance increases at specific pre-training settings were as high as 16.50 percentage points in F1-score, and relative gains were up to 21.15% in F1-score, respectively. These results demonstrate the effectiveness of the proposed hybrid model in handling dialectal ASA, outperforming previous approaches in key performance metrics.

Finally, we consider the observed class imbalance where the neutral class is underrepresented. To minimize the resulting bias, all metrics reported were calculated using weighted averages to

ensure that the performance of the minority class is accurately reflected. This approach, combined with the structure of our hybrid architecture, validated high performance despite the inherent heterogeneity of the dataset.

6 Conclusion

This study conducted an ablation study to evaluate the proposed hybrid DL model, AraBERT_CNN_MHA_BiLSTM, and to evaluate the contribution of each component to target ASA for the IAD. The proposed model was evaluated on two datasets: (i) the IAD dataset introduced by [10], and (ii) the IQAD31K dataset that was collected from Google Play Store application reviews in dialectal Arabic. Across both corpora, the model effectively captured fine-grained sentiment nuances in IAD. The complete model demonstrated a significant F1 score of 95.63% on “IQAD31K” and 94.50% on IAD datasets, outperforming the traditional ML methods like SVM and RF, which struggled with the complexity of the Arabic language. Additionally, the evaluation clearly demonstrates the role of CNN in local feature extraction. It highlights the importance of global context modeling and sequence in capturing deeper semantic relationships, particularly for ASA, where regional dialects present unique linguistic challenges.

The study outcomes indicate the importance of combining AraBERT’s contextual embeddings with CNN for local feature extraction, MHA for long-range dependencies, and BiLSTM for sequential modeling, developing a robust architecture designed adequately to the morphological and syntactic intricacies of Arabic, which collectively enhance feature extraction and classification accuracy for SA in the Iraqi dialect. These findings highlight the effectiveness of integrating these specific DL components, offering a structurally validated methodology for future ANLP studies, especially for underrepresented dialects. Future research will include comparative experiments utilizing modern, state-of-the-art **Large Language Models (LLMs)**, incorporating both global multilingual systems and advanced structures specifically designed for the Arabic language. This will assist us in identifying the performance gap and clarifying the present best practices in dialectal ASA.

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